

IPL Migration Considerations for PM43 and PC Series Printers



Introduction

IPL (Intermec Printer Language) is a printer command language that has been used in Intermec printers with customer applications for many years. Many applications have hard-coded IPL commands for printing with Intermec printers.

New Intermec printers introduce a completely new firmware architecture that takes advantage of modern technologies, adds features, improves usability and prepares for exciting future enhancements. This change in architecture includes a new version IPL that does not inherit source code from earlier implementations.

Backward compatibility with previous versions of IPL was a significant goal of the development; however changes in hardware, mechanics, and platform standards may result in differences in behavior, functionality and usage.

This document is intended to capture and discuss differences of new IPL compared to IPL on legacy printers. It will be updated from time to time to further clarify and reflect changes/additions in support.

Supported Printers

The IPL command language support discussed in this document is applicable to the following printers:

- PC23d, PC43d and PC43t
- PM43 and PM43c

Using this Document

The following topics are covered in this document:

- General Differences – System-level changes in the way the new printers work
- Barcode Differences – Changes in general barcode behavior and specific exceptions for certain barcodes
- Font Differences – Description of new font support and changes in font behavior
- Unsupported Commands – A complete list of IPL commands no longer supported in the new printers

- Supported Modes and Commands – A complete list of IPL commands that continue to be supported in the new printers
- New Commands – A complete list of new IPL commands introduced to support features in the new printers

General Differences

Page mode and Xon/Xoff with Status

Page mode and Xon/Xoff with Status are not supported in the initial releases of IPL for new printers. These features will be supported in the next official firmware release, scheduled for January, 2013.

RFID Support

RFID is not supported in the initial implementation. Support for RFID may be added in future versions of IPL.

Display Language Support

The following languages are supported in printers that have an LCD display:

- Standard: English, French, German, Italian, Portuguese, Russian, Spanish
- Optional: Simplified Chinese, Traditional Chinese, Korean

Label Printing

Labels are printed based on the size of the image buffer configured by setting the length and width of the label in the printer. The printer should be configured to accommodate or extend printing beyond one label.

Printing Position

The fixed offset of 3mm for all print heads along system x axis (IPL y axis) printing position may not be same as PD41/42, PF2/4i, PM4i and PX4/6i upgrade printers.

You may need to adjust system X margin (IPL y axis) or start/stop (IPL x axis) adjust to achieve legacy printing positions.

Print Speed

The default print speed for IPL is the system print speed and is dependent on the printer platform. There is not an IPL-specific print speed default.

Media Sensitivity and Darkness

The defaults for media sensitivity and darkness are driven by the system defaults and are dependent on the printer platform.

Printing Control Characters

IPL does not support printing control characters in rotated orientation for bitmap font c0, c1, c2 and c7.

In the following image the first line, rotated control character printing, is not supported.



Date and Time (d4/d5 in field data)

IPL prints the date format correctly according to the IPL manual, however the print outs are different than PF2/4i, PM4i and PX4/6i printers.

New Platform IPL:

2012/02/03

IPL on PF2/4i, PM4i, PX4/6i:

01/01/1980

IPL Command Reference Manual (d command)

If n = 4, m1 indicates how the date is formatted:

m1	Format for date
0	YY/MM/DD (default)
1	YYYY/MM/DD
2	DD/MM/YY
3	DD/MM/YYYY

Configuration Transmits

IPL utilizes the system start and stop adjust configuration for the following settings:

- <SI>f - Label rest point (Stop Adjust)
- <SI>F - Top of form (Start Adjust)
- <SI>X - Origin Adjust (Start Adjust)
- <SI>r - Label Retract distance (Start Adjust)

If you have configured these settings via command, the resulting value may not be accurate.

Error Handling/Error Codes

Error codes may be different when compared to PD41/42, PF2i/4i, PM4i and PX4i/6i printers.

Printer Webpage

- The printer webpage does not include an option to configure Emulation and Legacy modes
- User defined characters and fonts are not reflected in test labels printed from the webpage

Barcode Differences

- 1D and 2D barcodes may have different patterns, but content and scan results are the same
- IPL barcodes follow standard data type input - invalid data may result in unexpected behavior
- Control characters in barcode data are not supported.
- In some cases distance between the barcode and human readable is not the same as legacy printers

UPC/EAN

IPL barcode select command - c7

IPL application will print the 1st and last digit outside the guard bars for EAN/UPC barcodes as compared to PF2i/4i, PM4i and PX4i/6i upgrade printers.

The following data types are accepted:

Values: 0-9

CODABAR

IPL barcode select command - c4

The following data types are accepted:

No of chars: No restriction

Check char: No

Values: 0~9 - \$: / . +

Start chars: A,B,C,D

Stop chars: A,B,C,D

CODE11

IPL barcode select command - c5

The following data types are accepted:

No of chars: No restrict

Check char: No

Values: 0~9 - (minus)

Start chars: No

Stop chars: No

Font Differences

In 2010 Intermec moved to Monotype font technology with the PD41/42, PF2i/4i, PM4i and PX4i/6i upgrade printers. Why is the change to Monotype font technology important?

- Previous font support was based on out-dated font technology no longer supported by the vendor
- Added support for industry-standard TrueType and OpenType fonts (any font that follows these standards will work in the printers)
- Added support for global languages (not just characters) with complex text layout capabilities
- New printer resident fonts do not exactly match legacy fonts in terms of character look, but care was taken to ensure the new fonts are metrically compatible (fit in the same space) with legacy fonts

Legacy Mode

Under legacy mode, some fonts print longer when compared to PF2i/4i, PM4i and PX4i/6i upgrade printers but are similar in length to legacy printers. Only character spacing is supported.



Outline Fonts

Download of outline fonts using IPL commands is not supported.

C0 to c7 Fonts

When compared to PF2i/4i, PM4i and PX4i/6i upgrade printers fonts at 400 dpi may look different.

Unsupported Commands

- Most commands that were not carried over are due to changes in hardware and available technology
- Commands with an (^) are features that have been replaced by functionality in the printers system UI
- Commands with an (*) are likely out of use, but may be added to future releases if needed

Communications Commands

IPL Syntax	Command
<ESC>d *	Auto-Transmit 2, Enable
<ESC>e *	Auto-Transmit 3, Enable
<ESC>J *	RFID Tag, Read
<ESC>j *	Auto-Transmit 1, Enable
<ESC>k *	Auto-Transmit 1, 2, and 3, Disable
<SI>P *	Communications Port Configuration, Set
<ESC><SYN>	Message Delay, Set

Configuration Commands

IPL Syntax	Command
<SI>a	Audible Alarm, Enable/Disable
<SI>b	Takeup Motor Torque, Increase
<SI>Bs,ACTIVE,1	Activate Settings, Set
<SI>Bs,AUTH	Authenticate, Set
<SI>Bs,CONNECT	Connectable State, Set
<SI>Bs,ENCRYPT	Encryption, Set
<SI>Bs,FACTORYDEF,1	Factory Default, Set
<SI>Bt,ACTIVE	Activate Settings, Transmit
<SI>Bt,CONNECT	Connectable State, Transmit
<SI>Bt,ENCRYPT	Encryption, Transmit
<SI>Bt,FACTORYDEF	Factory default, Transmit
<SI>Bt,FIRMWARE	Firmware Version, Transmit
<SI>E	Display and Power Management Configuration, Set
<SI>H	Printhead Pressure, Set
<SI>I	Number of Image Bands, Set
<SI>i	IBM Language Translation, Enable/Disable
<SI>J *	RFID Parameters, Set
<SI>K *	RFID Tag Void Text, Set
<SI>N	Amount of Storage, Define
<SI>O	Online or Offline on Power-Up
<SI>P	Communication Port Configuration, Set
<SI>p	Pin 11/20 Protocol, Set
<SI>s	Interlabel Ribbon Save, Enable/ Disable
<SI>U	Printhead Test Parameters, Set
<SI>ws,ACTIVE,1	Activate Settings, Set (802.11 or Wireless LAN)
<SI>ws,AUTH	802.11 Authentication, Set
<SI>ws,WPA	WPA, Set

IPL Syntax	Command
<SI>ws,WPA2	WPA2, Set
<SI>wt,ACTIVE	Activate Settings, Transmit (802.11x or Wireless LAN)
<SI>wt,WPA	WPA, Transmit
<SI>wt,WPA2	WPA2, Transmit
<SI>xp	Change Password
<SI>Z	Ribbon Save Zones, Set
<SI>% *	Image Compression Set
<SYN>	Intercharacter Delay, Set

Immediate Commands

IPL Syntax	Command
<VT> *	Status Dump

Print Commands

IPL Syntax	Command
<ESC>Z	User-Defined Tables, Transmit
<ESC>%*	Image compression, Transmit

Program Mode Commands

IPL Syntax	Command
a *	RFID Tag Field Setup
C	Command Tables, Load
g	User Defined Characters and Graphics, Print (use U command)
g	Pitch Size, Set
J	Outline Font, Clear or Create
j	Outline Font, Download
N	Current Edit Session, Save
n *	RFID Tag Protect
p*	Pages, Print
Q *	RFID Tag Write Field, Create or Edit
t	User Defined Fonts, Print
v	Print Line Dot Count Limit, Set

Test and Service Commands

IPL Syntax	Command
A	Ambient Temperature, Transmit
B	Printhead Resistance Test, Begin
C	Pitch Label, Print
G ^	Transmissive Sensor Value, Transmit
L ^	Label Path Open Sensor Value, Transmit
M ^	Reflective Sensor Value, Transmit
p ^	Printhead Temperature Sensor Value, Transmit
S ^	Printhead Resistance Values, Transmit
T ^	Label Taken Sensor Value, Transmit
U ^	12 Volt Supply Value, Transmit
V ^	Printhead Volt Supply Value, Transmit

Supported Modes and Commands

The modes and commands listed in this section are supported in the new implementation of IPL. Please refer to the IPL Command Reference Manual for syntax and usage.

Supported Modes

IPL Mode	Definition
Advanced Mode	IPL default mode.
Program Mode	IPL mode to design bar code label formats.
Print Mode	IPL mode to download data to the printer and print bar code labels.
Emulation Mode	10 mil for 100 dpi printhead Data Shift One Bit per Byte
Legacy Mode	Font spacing
Test and Service Mode	Minimal Support: Enter command (<ESC>T) Exit command (R) Factory Default command (D) Test labels commands (h, Q, s)

Supported Commands

IPL Syntax	Description
;	Command Terminator
<ACK>	First Data Entry Field, Select
<BEL>	Error code, Request
<BS>	Warm Boot
<CAN>	Clear all data
<CR>	Next Data Entry Field, Select
	Clear Data From Current Field
<DLE>	Reset
	Abort Print Job
<ENQ>	Status Enquiry
<EOT>	Postamble, Set
<ESC><SP>	Start and Stop Codes (Code 39), Print
<ESC>C	Advanced Mode, Select
<ESC>c	Emulation Mode, Enter
<ESC>D	Field Decrement, Set
<ESC>E	Format, Select
<ESC>F	Field, Select
<ESC>G	Page, Select
<ESC>g	Direct Graphics Mode, Select
<ESC>H	Printhead Parameters, Transmit
<ESC>I	Field Increment, Set
<ESC>L	Label and Gap Length, Transmit
<ESC>M	Program Number, Transmit
<ESC>m	Memory Usage, Transmit
<ESC>N	Increment and Decrement, Disable
<ESC>O	Options Selected, Transmit
<ESC>P	Program Mode, Enter
<ESC>p	Configuration Parameters, Transmit

IPL Syntax	Description
<ESC>Q	Remaining Quantity and Batch Count, Transmit
<ESC>q	Date or Time, Transmit
<ESC>T	Test and Service Mode, Enter
<ESC>u	User-Defined Characters, Transmit (command format change)
<ESC>V	Battery Status and Information, Transmit
<ESC>v	Font, Transmit (command format change)
<ESC>x	Format, Transmit (command format change)
<ESC>.x	Execute shell commands
<ESC>y	Page, Transmit (command format change)
<ETB>	Print
<FF>	Form Feed
<FS>	Numeric Field Separator
<GS>	Alphanumeric Field Separator
<LF>	Command Terminator 2
<NUL>	Command Terminator 1
<RS>	Quantity Count, Set
<SI>A	Control Panel Access Permission, Set
<SI>Bs,DEV_NAME	Device Name, Set
<SI>Bs,DISCOVER	Discoverable State, Set
<SI>Bs,.PASSKEY	Passkey, Set
<SI>Bs,RESERVE,n	Reserve Printer, Set
<SI>Bt	All Bluetooth Settings, Transmit
<SI>Bt,AUTH	Authenticate, Transmit
<SI>Bt,DEV_ADDR	Device Address, Transmit
<SI>Bt,DEV_NAME	Device Name, Transmit
<SI>Bt,DISCOVER	Discoverable State, Transmit
<SI>Bt,.PASSKEY	Passkey, Transmit
<SI>Bt,RESERVE	Reserve Printer, Transmit
<SI>C	Emulation or Advanced Mode on Power-Up
<SI>c	Cutter, Enable or Disable
<SI>D	End-of-Print Skip Distance, Set. Default is '0' regardless of media loaded
<SI>d	Dark Adjust, Set
<SI>e	Media Fault Recovery Mode, Set
<SI>F	Top of form, Set
<SI>f	Label Rest Point, Adjust
<SI>g	Media Sensitivity, Select
<SI>h	Printhead Loading Mode, Select
<SI>L	Maximum Label Length, Set
<SI>I	Printer Language, Select
<SI>n	Setting Network Parameters in Manual Mode

IPL Syntax	Description
<SI>o	Direct Graphics Emulation Mode, Enable or Disable
<SI>P	Communication Port Configuration, Set
<SI>q	Date or Time, Set
<SI>R	Label Retract, Enable or Disable retract. Enable/ disable n=2 unsupported
<SI>r	Label Retract Distance, Set
<SI>S	Print Speed, Set
<SI>T	Label Stock Type, Select
<SI>t	Self-Strip, Enable or Disable
<SI>V	Battery Low Voltage Threshold, Set
<SI>W	Label Width, Set
<SI>ws,ACQUIRE_PAC	Acquire PAC, Set
<SI>ws,CA_CERT	Certificate Authority, Set
<SI>ws,CHANNEL	Active Channel, Set
<SI>ws,EAP_PASS	EAP Password, Set
<SI>ws,EAP_TYPE	EAP Type, Set
<SI>ws,.EAP_USER	EAP User Name, Set
<SI>ws,INNER_AUTH	Inner Authentication, Set
<SI>ws,MODE	Mode, Set
<SI>ws,PAC	Protected Access Credential, Set
<SI>ws,PWR_MODE	Power Management Mode, Set
<SI>ws,ROAM	Roaming Bias, Set
<SI>ws,.SERVER_CN1	Server Common Name 1, Set
<SI>ws,.SERVER_CN2	Server Common Name 2, Set
<SI>ws,SSID	SSID (Network Name), Set
<SI>ws,TTLS_USER	TTLS User Name, Set
<SI>ws,VALIDATE	Validate, Set
<SI>ws,WEP1	WEP, Set
<SI>ws,WEP2	WEP, Set
<SI>ws,WEP3	WEP, Set
<SI>ws,WEP4	WEP, Set
<SI>ws,.WEP_KEY	WEP Key, Set
<SI>ws,WPA_PSK	WPA Pre-Shared Key, Set
<SI>wt	All settings, Transmit (Wireless LAN or 802.11x)
<SI>wt,ACQUIRE_PAC	Acquire PAC, Transmit
<SI>wt,AP_MAC	Access Point Information, Transmit
<SI>wt,AUTH	802.11 Authentication, Transmit
<SI>wt,CA_CERT	Certificate Authority, Transmit
<SI>wt,CHANNEL	Active chaneel, Transmit
<SI>wt,EAP_PASS	EAP Password, Transmit
<SI>wt,EAP_TYPE	EAP Type, Transmit
<SI>wt,.EAP_USER	EAP User Name, Transmit
<SI>wt,INNER_AUTH	Inner Authentication, Transmit
<SI>wt,MODE	Mode, Transmit
<SI>wt,PAC	Protected Access Credential, Transmit

IPL Syntax	Description
<SI>wt,PWR_MODE	Power Management Mode, Transmit
<SI>wt,.REGION	Region, Transmit
<SI>wt,ROAM	Roaming Bias, Transmit
<SI>wt,.SERVER_CN1	Server Common Name 1, Transmit
<SI>wt,.SERVER_CN2	Server Common Name 2, Transmit
<SI>wt,SIGNAL	Signal Strength, Transmit
<SI>wt,SPEED	Speed, Transmit
<SI>wt,SSID	SSID (Network Name), Transmit
<SI>wt,STATE	State, Transmit
<SI>wt,TTLS_USER	TTLS User Name, Transmit
<SI>wt,VALIDATE	Validate, Transmit
<SI>wt,WEP1	WEP, Transmit
<SI>wt,WEP2	WEP, Transmit
<SI>wt,WEP3	WEP, Transmit
<SI>wt,WEP4	WEP, Transmit
<SI>wt,.WEP_KEY	WEP Key, Transmit
<SI>wt,WPA_PSK	WPA Pre-Shared Key, Transmit
<SI>X	Label Origin, X-Y Adjust
<SI>xu	Change User
<SI>z	Slash Zero, Enable or Disable
<SO>	Cut
<SOH>	Preamble, Set
<SUB>	Data Shift - International Characters
<US>	Batch Count, Set
A or F	Format, Create or Edit
B	Bar Code Field, Create or Edit
B	Border Around Human-Readable Text, Define
C	Bar Code, Select Type
C	Font Type, Select
C	Graphic, Select
D	Factory Defaults, Reset
D	Field, Delete
d	Field Data, Define Source
E	Format, Erase
e	Data Source for Format in a Page, Define
f	Field Direction, Define
f	Formats, Print
G	User-Defined Character, Clear or Create
H	Human Readable Field, Create or Edit
h	Hardware Configuration Label, Print
h	Height Magnification of Bar, Box, or UDC, Define Range: Box - 1 to 6000 Range: Human readable/ graphics - 1 to 250

IPL Syntax	Description
I	Interpretive Field, Edit
i	Interpretive Field, Enable or Disable
K	Dark Adjust
k	Point Size, Set
L	Line Field, Create or Edit
l	Length of Line or Box Field, Define
M	Format Position in a Page, Assign
m	Format Position From Page, Delete
m	Text Box, Set Margins
n	RFID Tag Protect
O	Format Offset Within a Page, Define
o	Field Origin, Define
p	Code 39 Prefix Character, Define
Q	Print Quality Label, Print
q	Format Direction in a Page, Define
R	Program Mode, Exit
R	Test and Service Mode, Exit
r	Character Rotation or Bar Code Ratio, Define
S	Page, Create or Edit
s	Page, Delete
s	Software Configuration Label, Print
T	Bitmap User Defined Font, Clear or Define

IPL Syntax	Description
t	User Defined Font Character, Create
U	User Defined Character Field, Create or Edit
u	Graphic or UDC, Define
W	Box Field, Create or Edit
w	Width of Line, Box, Bar, or Character, Define Range: Box – 1 to 6000 Range: Human readable/ graphics – 1 to 250
X	Character Bitmap Origin Offset, Define
x	Bitmap Cell Width for Graphic or UDF, Define
x	Text Box, Create
y	Bitmap Cell Height for Graphic or UDF, Define
Z	Font Character Width, Define
z	Intercharacter Space for UDF, Define
0x21	Direct Graphics Mode, Change Origin
0x22	Direct Graphics Mode, End of line
0x24	Direct Graphics Mode, Repeat Last Line
0x25	Direct Graphics Mode, Transition Black
0x26	Direct Graphics Mode, Transition White
0x27	Direct Graphics Mode, Raw Bitmap Data Follows
0x28	Direct Graphics Mode, End of Bitmap

New Commands

Wireless 802.11 Commands

The table below lists commands to set printer parameters for Wireless 802.11 communications. Both legacy and new commands are supported together, except for the few legacy commands labeled as unsupported in table.

Legacy Command	Legacy Command Parameters/ Response	New Command	New Command Parameters/ Response
<SI>ws,ACQUIRE_PAC	Acquire PAC, Set Parameters: ON OFF	<SI>ws,ACQUIRE PAC	Acquire PAC, Set Parameters: ON OFF
<SI>ws,AUTH Unsupported	NA	<SI>ws,SECURITY TYPE	802.11 Security Type, Set Parameters: None WEP (static) Pre-shared key PEAP TLS TTLS LEAP EAP-FAST
<SI>ws,CA_CERT,n	Certificate Authority, Set Parameters: n (CA Certificate Path Value)	<SI>ws,CA CERTIFICATE,n	Certificate Authority, Set Parameters: n (CA Certificate Path Value)
<SI>ws,CHANNEL,n	Active Channel, Set Parameters: n(Numeric values 1 to 14)	<SI>ws,CHANNEL,n	Active Channel, Set Parameters: n(Numeric values 1 to 14)
<SI>ws,ACQUIRE_PAC	Acquire PAC, Set Parameters: ON OFF	<SI>ws,ACQUIRE PAC	Acquire PAC, Set Parameters: ON OFF
<SI>ws,AUTH Unsupported	NA	<SI>ws,SECURITY TYPE	802.11 Security Type, Set Parameters: None WEP (static) Pre-shared key PEAP TLS TTLS LEAP EAP-FAST
<SI>ws,CA_CERT,n	Certificate Authority, Set Parameters: n (CA Certificate Path Value)	<SI>ws,CA CERTIFICATE,n	Certificate Authority, Set Parameters: n (CA Certificate Path Value)
<SI>ws,CHANNEL,n	Active Channel, Set Parameters: n(Numeric values 1 to 14)	<SI>ws,CHANNEL,n	Active Channel, Set Parameters: n(Numeric values 1 to 14)
<SI>ws,EAP_PASS,n	User Password, Set Parameters: n (It can be 96 Characters)	<SI>ws,USER PASSWORD,n	User Password, Set Parameters: n (It can be 96 Characters)
<SI>ws,EAP_TYPE	Association, Set Parameters: OPEN/WEP SHARED/WEP WPA/TKIP WPA2/AES	<SI>ws,ASSOCIATION	Association, Set Parameters: OPEN/WEP SHARED/WEP WPA/TKIP WPA2/AES
<SI>ws,.EAP_USER,n	User Name, Set Parameters: n (It can be 96 Characters)	<SI>ws,USER NAME,n	User Name, Set Parameters: n (It can be 96 Characters)

Legacy Command	Legacy Command Parameters/ Response	New Command	New Command Parameters/ Response
<SI>ws,INNER_AUTH	Inner Authentication, Set Parameters: PAP MSCHAPv2 EAP/MSCHAPv2 EAP/MD5 EAP/GTC EAP/TLS	<SI>ws,INNER AUTHENTICATION	Inner Authentication, Set Parameters: PAP MSCHAPv2 EAP/MSCHAPv2 EAP/MD5 EAP/GTC EAP/TLS
<SI>ws,MODE	Mode, Set Parameters: ADHOC INFRA	<SI>ws,NETWORK TYPE	Mode, Set Parameters: Ad Hoc Infrastructure
<SI>ws,PAC	Protected Access Credential, Set Parameters: PAC Path value	<SI>ws,PAC	Protected Access Credential, Set Parameters: PAC Path value
<SI>ws,PWR_MODE	Power Management Mode, Set Parameters: 0 1 2	<SI>ws,POWER MODE	Power Management Mode, Set Parameters: Constant Awake Mode Power Saving Fast Power Saving
<SI>ws,ROAM	Roaming Bias, Set Parameters: 0 1 2 3	<SI>ws,ROAMING	Roaming Bias, Set Parameters: Off Level 1 Level 2 Level 3
<SI>ws,,SERVER_CN1,n	Server Common Name 1, Set Parameters: n (It can be up to 96 Characters)	<SI>ws,,SERVER COMMON NAME 1,n	Server Common Name 1, Set Parameters: n (It can be up to 96 Characters)
<SI>ws,,SERVER_CN2,n	Server Common Name 2, Set Parameters: n (It can be up to 96 Characters)	<SI>ws,,SERVER COMMON NAME 2,n	Server Common Name 2, Set Parameters: n (It can be up to 96 Characters)
<SI>ws,SSID	SSID (Network Name), Set Parameters: n (0 to 32 octets)	<SI>ws,NETWORK NAME (SSID)	SSID (Network Name), Set Parameters: n (0 to 32 octets)
None	NA	<SI>ws,HIDDEN SSID	Hidden SSID, Set Parameters: ENABLE DISABLE
<SI>ws,TTLS_USER,n	Anonymous Name, Set Parameters: n (up to 96 Characters)	<SI>ws,ANONYMOUS NAME,n	Anonymous Name, Set Parameters: n (up to 96 Characters)
<SI>ws,VALIDATE	Validate, Set Parameters: ON OFF	<SI>ws,VALIDATE CERTIFICATE	Validate, Set Parameters: ON OFF
<SI>ws,WEP1,n	Network key, Set Parameters: n (WEP key value)	<SI>ws,NETWORK KEY 1	Network key, Set Parameters: n (WEP key value)
<SI>ws,WEP2,n	Network key, Set Parameters: n (WEP key value)	<SI>ws,NETWORK KEY 2	Network key, Set Parameters: n (WEP key value)
<SI>ws,WEP3,n	Network key, Set Parameters: n (WEP key value)	<SI>ws,NETWORK KEY 3	Network key, Set Parameters: n (WEP key value)

Legacy Command	Legacy Command Parameters/ Response	New Command	New Command Parameters/ Response
<SI>ws,WEP4,n	Network key, Set Parameters: n (WEP key value)	<SI>ws,NETWORK KEY 4	Network key, Set Parameters: n (WEP key value)
<SI>ws,,WEP_KEY	Network Key Index, Set Parameters: 1 2 3 4	<SI>ws,NETWORK KEY INDEX	Network Key Index, Set Parameters: 1 2 3 4
None	NA	<SI>ws,CLIENT KEY	Client Key, Set Parameters: n (up to 128 Characters)
<SI>ws,WPA_PSK,n	Pre-Shared Key, Set Parameters: n (8 to 64 characters long)	<SI>ws,PRE-SHARED KEY,n	Pre-Shared Key, Set Parameters: n (8 to 64 characters long)
<SI>wt	All settings, Transmit (Wireless LAN or 802.11x)	<SI>wt	All settings, Transmit (Wireless LAN or 802.11x)
<SI>wt,ACQUIRE_PAC	Acquire PAC, Transmit	<SI>wt,ACQUIRE PAC	Acquire PAC, Transmit
<SI>wt,AP_MAC	Access Point Information, Transmit	<SI>wt,ACCESS POINT MAC	Access Point Information, Transmit
<SI>wt,AUTH	802.11 Authentication, Transmit	<SI>wt,SECURITY TYPE	802.11 Authentication, Transmit
<SI>wt,CA_CERT	Certificate Authority, Transmit	<SI>wt,CA CERTIFICATE	Certificate Authority, Transmit
<SI>wt,CHANNEL	Active channel, Transmit	<SI>wt,CHANNEL	Active channel, Transmit
<SI>wt,EAP_PASS	EAP Password, Transmit	<SI>wt,USER PASSWORD	User Password, Transmit
<SI>wt,EAP_TYPE	EAP Type, Transmit	<SI>wt,ASSOCIATION	ASSOCIATION, Transmit
<SI>wt,,EAP_USER	EAP User Name, Transmit	<SI>wt,,USER NAME	User Name, Transmit
<SI>wt,INNER_AUTH	Inner Authentication, Transmit	<SI>wt,INNER AUTHENTICATION	Inner Authentication, Transmit
<SI>wt,MODE	Mode, Transmit	<SI>wt,NETWORK TYPE	Network Type, Transmit
<SI>wt,PAC	Protected Access Credential, Transmit	<SI>wt,PAC	Protected Access Credential, Transmit
<SI>wt,PWR_MODE	Power Management Mode, Transmit	<SI>wt,POWER MODE	Power Management Mode, Transmit
<SI>wt,,REGION	Region, Transmit	<SI>wt,,REGION	Region, Transmit
<SI>wt,ROAM	Roaming Bias, Transmit	<SI>wt,ROAMING	Roaming Bias, Transmit
<SI>wt,,SERVER_CN1	Server Common Name 1, Transmit	<SI>wt,,SERVER COMMON NAME 1	Server Common Name 1, Transmit
<SI>wt,,SERVER_CN2	Server Common Name 2, Transmit	<SI>wt,,SERVER COMMON NAME 2	Server Common Name 2, Transmit
<SI>wt,SIGNAL	Signal Strength, Transmit	<SI>wt,SIGNAL	Signal Strength, Transmit
<SI>wt,SPEED	Speed, Transmit	<SI>wt,SPEED	Speed, Transmit
<SI>wt,SSID	SSID (Network Name), Transmit	<SI>wt,NETWORK NAME (SSID)	SSID (Network Name), Transmit
None	NA	<SI>wt,HIDDEN SSID	Hidden SSID access, Transmit
<SI>wt,STATE	State, Transmit	<SI>wt,STATE	State, Transmit
<SI>wt,TTLS_USER	TTLS User Name, Transmit	<SI>wt,ANONYMOUS NAME	ANONYMOUS NAME, Transmit
<SI>wt,VALIDATE	Validate, Transmit	<SI>wt,VALIDATE CERTIFICATE	Validate, Transmit
<SI>wt,WEP1	WEP, Transmit	<SI>wt,NETWORK KEY 1	NETWORK KEY 1, Transmit
<SI>wt,WEP2	WEP, Transmit	<SI>wt,NETWORK KEY 2	NETWORK KEY 2, Transmit
<SI>wt,WEP3	WEP, Transmit	<SI>wt,NETWORK KEY 3	NETWORK KEY 3, Transmit
<SI>wt,WEP4	WEP, Transmit	<SI>wt,NETWORK KEY 4	NETWORK KEY 4, Transmit
<SI>wt,,WEP_KEY	WEP Key, Transmit	<SI>wt,NETWORK KEY INDEX	NETWORK KEY INDEX, Transmit
<SI>wt,WPA_PSK	WPA Pre-Shared Key, Transmit	<SI>wt,PRE-SHARED KEY	Pre-Shared Key, Transmit
None	NA	<SI>wt,CLIENT KEY	Client Key, Transmit

Bluetooth Commands

The table below lists legacy and new commands supported in IPL with relevant parameters. Both legacy and new commands are supported together, except for the few legacy commands labeled as unsupported in table.

Legacy Command	Legacy Command Parameters/ Response	New Command	New Command Parameters/ Response
<SI>Bs,AUTH,n Unsupported	NA	<SI>Bs,SECURITY,s	Security, Set Parameters (n): Disable Auth Auth and Encrypt
None	None	<SI>Bs,POWER SAVING MODE	Power saving, setting Parameter Enable Disable
<SI>Bs,DEV_NAME,s...s	Device Name, Set Parameters: s (16 Printable Characters)	<SI>Bs,DEVICE NAME,s	Device Name, Set Parameters: s (16 Printable Characters)
<SI>Bs,DISCOVER,n	Discoverable State, Set Parameters: 0 (Disable) 1 (Enable)	<SI>Bs,DISCOVER,s	Discoverable State, Set Parameters: Disable Enable
<SI>Bs,ENCRYPT,n Unsupported	NA	None	None
<SI>Bs,.PASSKEY,s...s	Passkey, Set Parameters: s (16 Printable Characters)	<SI>Bs,PASS KEY,s	Passkey, Set Parameters: s (16 Printable Characters)
<SI>Bs,RESERVE,n	Reserve Printer, Set Parameters: 0 (Disable reserving) 1 (Enable reserving)	<SI>Bs,RESERVE,s	Reserve Printer, Set Parameters: Disable Enable
<SI>Bt	All Bluetooth Settings, Transmit	<SI>Bt	All Bluetooth Settings, Transmit
<SI>Bt,AUTH	Security, Transmit	<SI>Bt,SECURITY	Security, Transmit
<SI>Bt,DEV_ADDR	Device Address, Transmit	<SI>Bt,DEVICE ADDRESS	Device Address, Transmit
<SI>Bt,DEV_NAME	Device Name, Transmit	<SI>Bt,DEVICE NAME	Device Name, Transmit
<SI>Bt,DISCOVER	Discoverable State, Transmit	<SI>Bt,DISCOVER	Discoverable State, Transmit
<SI>Bt,ENCRYPT Unsupported	NA	None	None
<SI>Bt,.PASSKEY	Passkey, Transmit	<SI>Bt,.PASS KEY	Passkey, Transmit
<SI>Bt,RESERVE	Reserve Printer, Transmit	<SI>Bt,RESERVE	Reserve Printer, Transmit

Technical Support

Should you experience issues migrating from legacy printers to newer printers with IPL, a Knowledge Central case should be opened with Intermec technical support – the fastest path to resolution.

Email: KCSupport@custhelp.com

Web: <http://intermec.custhelp.com/app/home>

Conclusion

Understanding the core differences between legacy and new printers can help set expectations and provide a smoother transition and better overall printing experience.

Intermec's new printers introduce a completely new firmware architecture that takes advantage of new technologies, adds features, improves usability and prepares for exciting future enhancements. This change in architecture includes a new version of IPL that does not inherit source code from earlier implementations.

Considerable care was taken to ensure backward compatibility with previous versions of IPL:

- 73% of all IPL commands were carried over from legacy IPL implementations
- 70 commands were added to enhance wireless security and Bluetooth configuration
- Over 7,000 known IPL use cases were tested before the release of IPL
- Most commands that were not carried over are due to changes in hardware and available technology
- Some commands, infrequently used or considered to be out of use, may be added to future development as use cases become available

Intermec is committed to providing you with the best possible experience and we are confident that Intermec and our new printers will exceed your expectations. Should you have difficulty migrating from legacy to new printers please contact us – Intermec will make it right.

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